From Draft Capital to Fantasy Output: Predicting Rookie Performance with Machine Learning

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***Abstract*—This study evaluates the use of statistical machine learning methods to model rookie fantasy football performance using pre-draft and college-level data. Two prediction tasks are considered: estimating total rookie season PPR fantasy points and classifying whether a player is fantasy relevant based on position-specific thresholds. Multiple supervised learning models were compared, including linear regression, logistic regression, decision trees, random forests, and neural networks. Random Forest produced the strongest overall performance, achieving a classification ROC-AUC of approximately 66% and the lowest regression error with an RMSE of approximately 94 points. Neural networks showed clear signs of overfitting and did not outperform simpler models, highlighting the limitations of complex models on smaller structured datasets. Feature importance analysis revealed that draft capital is the dominant driver of total fantasy production, while college performance metrics such as yards and touchdowns play a larger role in determining fantasy relevance. Unsupervised learning methods further showed that rookie players naturally group into distinct archetypes with different risk and production profiles. The model was also applied to generate forward-looking predictions for the 2026 rookie class, illustrating how historical patterns translate into probabilistic outcomes rather than deterministic forecasts. These results emphasize that model selection should be driven by data characteristics and that even relatively simple models can provide meaningful insights for fantasy football decision-making.**

I. Introduction

Predicting rookie performance in fantasy football is one of the most uncertain problems in sports analytics. Unlike veteran players, rookies enter the league without any NFL-level production, so projections rely heavily on indirect signals such as draft capital, college performance, and physical attributes. Because of this, outcomes can vary widely. Some players produce immediately and become fantasy assets, while others struggle to earn playing time or fail to meet expectations.

From a practical standpoint, this problem matters a lot in fantasy football, especially in formats where rookie production can shift the outcome of a season. Being able to identify which rookies will actually contribute meaningful points provides a clear competitive advantage. However, the challenge comes from the fact that the relationships between draft position, production, and opportunity are not simple or linear.

This study approaches the problem using statistical machine learning methods with the goal of evaluating how well different models can predict rookie outcomes using structured pre-draft data. Two prediction tasks are considered.

The first task is a regression problem, where the objective is to estimate total rookie season PPR fantasy points. This provides a direct measure of overall fantasy production.

The second task is a classification problem, where the goal is to determine whether a player is fantasy relevant. Fantasy relevance is defined using position-specific thresholds based on typical roster and scoring structures. A quarterback is considered relevant if they finish within the top 12 at the position. Running backs are considered relevant if they finish within the top 24. Wide receivers are considered relevant within the top 36, and tight ends within the top 12. These thresholds reflect common league formats and provide a realistic benchmark for meaningful contribution.

To solve these tasks, multiple supervised learning models are implemented and compared. Linear regression and logistic regression are used as baseline models to establish a reference point. Decision trees and random forests are then applied to capture nonlinear relationships and interactions between features. Neural networks are also included to explore more flexible modeling approaches, with additional analysis on how training behavior changes when early stopping is introduced.

In addition to predictive modeling, this study also focuses on understanding the structure of the data. Feature importance analysis is used to identify which variables contribute most to predictions, helping explain why certain models perform better. Unsupervised learning techniques such as Principal Component Analysis (PCA) and KMeans clustering are also applied to explore whether players naturally group into different archetypes based on their characteristics.

The primary goal of this project is not to deploy a finalized prediction system, but rather to evaluate how well different statistical learning methods can model rookie performance using structured data. By comparing multiple modeling approaches across both regression and classification tasks, this study focuses on identifying which methods generalize best and what features most strongly influence outcomes. This evaluation-based approach provides a foundation for future predictive applications while offering insight into the strengths and limitations of each modeling technique.

II. Problem Formulation

The goal of this study is to model rookie fantasy football performance as a supervised learning problem using pre-draft and college-level player data. Each player is represented by a set of input features, along with corresponding target variables that describe their rookie season outcome.

Let $X \in \mathbb{R}^p$ denote the feature vector for a given player. These features include draft-related information, physical attributes, and college production statistics. In this study, the feature set consists of variables such as overall draft pick, age at draft, height, weight, total college yards, total college touchdowns, and games played. Categorical features, such as player position, are incorporated using encoding techniques.

Two prediction tasks are defined:

The first task is a regression problem, where the goal is to estimate total rookie season PPR fantasy points:

$$Y_{\text{reg}} = \text{Rookie Total PPR Fantasy Points}$$

The objective is to learn a function $f(X)$ such that:

$$\hat{Y}_{\text{reg}} = f(X)$$

approximates true rookie production as closely as possible.

The second task is a binary classification problem, where the goal is to determine whether a player is fantasy relevant:

$$Y_{\text{clf}} = \begin{cases} 1, & \text{if player is fantasy relevant} \\ 0, & \text{otherwise} \end{cases}$$

A model is trained to estimate the probability:

$$P(Y_{\text{clf}} = 1 \mid X)$$

and assign a class label based on this probability.

Fantasy relevance is defined using position-specific thresholds. A quarterback is considered relevant if they finish within the top 12 at the position. Running backs are considered relevant within the top 24, wide receivers within the top 36, and tight ends within the top 12. These thresholds reflect typical fantasy roster structures and provide a practical definition of meaningful contribution.

Both tasks are trained on historical rookie data and evaluated on unseen players. Regression performance is measured using error-based metrics, while classification performance is evaluated using accuracy and probability-based metrics.

This formulation enables a direct comparison between different statistical learning methods, focusing on their predictive accuracy and ability to generalize to unseen rookie classes. Rather than producing final deployment predictions, the emphasis of this study is on evaluating model performance across tasks and identifying which approaches are most effective given the structure of the data.

III. Data

The dataset used in this study was constructed manually using publicly available player data, primarily sourced from Pro Football Reference. The focus was on first-round offensive skill position players across multiple draft classes, with each observation representing a single player and their corresponding rookie season.

Specifically, the dataset includes all first-round quarterbacks, running backs, wide receivers, and tight ends drafted between 2010 and 2026. This range was chosen to provide a sufficient number of observations for model training while still reflecting modern NFL trends and usage patterns.

Data collection involved compiling draft information, physical attributes, and college production statistics for each player. Key features include overall draft pick, age at draft, height, weight, total college yards, total college touchdowns, and total games played. Positional information was also included as a categorical variable to capture differences between quarterbacks, running backs, wide receivers, and tight ends.

Two target variables were defined for this analysis. The first target corresponds to a regression task and represents total PPR fantasy points scored during a player's rookie season. The second target is a binary classification variable indicating whether a player was fantasy relevant, based on position-specific thresholds for meaningful production.

Once the dataset was compiled, it was organized into structured tables and exported into a single file for processing. Data validation was performed to ensure consistency across all observations. This included checking for missing values, verifying numerical ranges for features such as draft position and physical measurements, and confirming that all target variables were correctly aligned with their corresponding player records.

In addition, exploratory data analysis was conducted to better understand the distribution of key variables and identify any potential anomalies. This included examining the distribution of rookie fantasy points, checking class balance for the classification target, and confirming that feature values fell within reasonable ranges. These steps helped ensure that the dataset was suitable for modeling and did not contain obvious data quality issues.

Before applying machine learning models, preprocessing steps were performed to prepare the data. Numerical features were standardized to ensure consistent scaling, while categorical features such as position were encoded using one-hot encoding. The dataset was then split into training and testing sets to evaluate model performance on unseen data, with a fixed random seed used to ensure reproducibility.

The dataset combines multiple sources of player information into a structured format that supports both regression and classification modeling tasks, while maintaining consistency and data quality through validation and preprocessing.

The dataset was split temporally, with players drafted between 2010 and 2022 used for training, 2023 to 2025 used for testing, and the 2026 draft class reserved for future prediction.

IV. Methods

This study follows a structured modeling approach to evaluate how different statistical learning methods perform when predicting rookie fantasy football outcomes. The methodology is divided into multiple stages, starting with baseline models and progressing toward more complex approaches, followed by unsupervised learning techniques for additional data insight.

A. Baseline Models

The first stage of the analysis focuses on establishing baseline performance using simple linear models. These models serve as a reference point to evaluate whether more complex approaches provide meaningful improvements.

For the regression task, Linear Regression is used to model the relationship between input features and total rookie PPR fantasy points. The model assumes a linear relationship of the form:

$$\hat{Y}_{\text{reg}} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p$$

where x_i represents each feature and β_i are the learned coefficients. The model is trained by minimizing mean squared error, which penalizes larger prediction errors more heavily.

This approach provides a simple way to understand how features such as draft position or college production contribute to expected rookie output. However, it assumes that all relationships are linear, which may not hold in a real-world setting where interactions between variables are likely.

For the classification task, Logistic Regression is used to estimate the probability that a player is fantasy relevant. Instead of predicting a continuous value, the model outputs a probability using the logistic function:

$$P(Y_{\text{clf}} = 1 \mid X) = \frac{1}{1 + e^{-(\beta_0 + \beta^T X)}}$$

A threshold is then applied to assign a class label.

Logistic Regression is particularly useful as a baseline because it provides interpretable coefficients and establishes a clear benchmark for classification performance. It allows us to see how well a simple linear decision boundary can separate fantasy-relevant players from non-relevant ones.

These baseline models are intentionally simple. They help answer an important question early in the analysis, whether rookie performance can be explained by straightforward linear relationships, or if more flexible models are needed to capture underlying patterns in the data.

B. Tree-Based Models

To move beyond the limitations of linear models, tree-based methods were introduced to capture nonlinear relationships and feature interactions within the dataset. In the context of rookie performance, relationships between variables such as draft position, college production, and physical attributes are not expected to follow simple linear patterns. Decision trees and random forests provide a way to model these more complex structures.

Decision Tree models were first applied to both regression and classification tasks. A decision tree works by recursively partitioning the feature space into smaller regions based on feature thresholds, effectively creating a set of decision rules. For example, a split might separate players drafted within the top 10 from those drafted later, or distinguish players with high college production from those with lower output. This allows the model to capture nonlinear decision boundaries without requiring explicit feature transformations.

However, decision trees are known to be highly sensitive to the training data and can easily overfit, especially when the dataset is relatively small. This instability can lead to inconsistent performance across different splits of the data and reduces their reliability when applied to unseen players.

To address this limitation, Random Forest models were introduced. A random forest is an ensemble method that builds multiple decision trees using different subsets of the data and features, and then aggregates their predictions. For regression, predictions are averaged across trees, while for classification, a majority vote is used. This approach reduces variance and improves generalization by smoothing out the behavior of individual trees.

In this study, random forests were expected to perform well due to the structured nature of the dataset and the presence of feature interactions. Variables such as draft capital, production metrics, and positional indicators likely interact in ways that are difficult for linear models to capture but are naturally handled by tree-based methods.

The results confirmed this expectation, as random forests consistently produced stronger and more stable performance compared to both decision trees and baseline linear models across both regression and classification tasks. These performance differences are explored in detail in the Results section, where tree-based models are compared directly against baseline and neural network approaches.

C. Neural Network Models

Neural networks were implemented to evaluate whether a more flexible, nonlinear modeling approach could outperform traditional methods. A feedforward neural network architecture was used for both regression and classification tasks, consisting of multiple fully connected layers with nonlinear activation functions.

Two variations of the model were tested. The first model was trained without any constraints, allowing it to run for a fixed number of epochs. The second model incorporated early stopping, where training was halted once validation loss stopped improving. This was done to better understand how overfitting impacts performance on a dataset of this size.

The difference between the two approaches became clear when analyzing the training behavior. The unconstrained model showed a consistent decrease in training loss, while validation loss increased over time. This is a classic sign of overfitting, where the model begins to memorize the training data instead of learning patterns that generalize to new players.

When early stopping was applied, the overfitting behavior was reduced. Validation loss stabilized and no longer diverged from training loss. However, this improvement in training stability did not translate into better predictive performance.

This tradeoff was one of the more important findings in the study. While neural networks are powerful and flexible, they require a large amount of data to perform well. In this case, the dataset was relatively small, 164 players tracked, which limited the model's ability to generalize effectively. As a result, neural

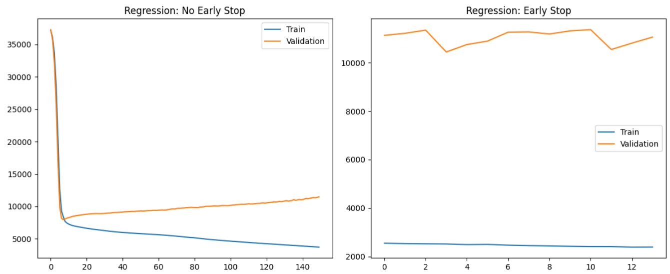

Fig. 1. Neural network training and validation loss comparing models with and without early stopping, highlighting overfitting behavior in the unrestricted model

networks did not outperform simpler models such as random forests.

This section highlights that model complexity does not guarantee better results. In structured datasets with limited sample size, simpler models can often provide more reliable and interpretable performance.

D. Model Evaluation

To evaluate model performance, separate metrics were used for the regression and classification tasks, since each problem requires a different way of measuring success.

For the regression task, the goal is to predict total rookie PPR fantasy points as accurately as possible. Two error-based metrics were used: Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). MAE provides a straightforward interpretation of the average prediction error, while RMSE places a larger penalty on larger mistakes. This distinction was important in this context, since large prediction errors, such as overestimating a player who produces very little, can significantly impact fantasy decision-making.

For the classification task, the objective is to determine whether a player is fantasy relevant. Because this is a binary outcome, multiple evaluation metrics were considered to capture different aspects of performance. Accuracy was used as a general baseline, but additional metrics including precision, recall, F1 score, and ROC-AUC were also included.

Precision and recall were particularly useful in understanding model behavior. Precision reflects how often predicted relevant players actually produced meaningful fantasy output, while recall measures how well the model identifies all relevant players. The F1 score balances these two, providing a single measure of overall classification effectiveness. ROC-AUC was used as the primary comparison metric across models, since it evaluates how well the model separates relevant and non-relevant players across all probability thresholds.

All models were trained on a training dataset and evaluated on a separate testing dataset. This separation ensures that performance metrics reflect how well each model generalizes to unseen rookie players, rather than how well it memorizes historical data.

It is important to note that the objective of this evaluation is not to generate finalized predictions for future draft classes, but to assess how well each model performs when applied to unseen data. This distinction ensures that model comparisons are based on generalization ability rather than memorization or overfitting.

E. Unsupervised Learning

In addition to supervised models, unsupervised learning techniques were applied to better understand the underlying structure of the dataset. While supervised models focus on prediction, unsupervised methods provide insight into how players naturally group based on their characteristics without using target labels.

Principal Component Analysis (PCA) was used to reduce the dimensionality of the feature space. Since the dataset contains multiple correlated features, PCA projects the data into a lower-dimensional space while preserving as much variance as possible. This makes it easier to visualize player distributions and identify patterns that are not immediately obvious in the original feature space.

The PCA visualization (Fig. 7) revealed clear separation between player positions, particularly with quarterbacks forming a distinct cluster compared to other skill positions. This indicates that positional differences are strongly represented in the feature space and likely contribute to differences in model behavior.

Following PCA, KMeans clustering was applied to group players into clusters based on their attributes. The goal was to determine whether players naturally fall into distinct archetypes without using any label information.

The clustering results, also shown in Fig. 7, indicate that players can be grouped into three primary clusters, each with different characteristics in terms of production and fantasy relevance. One cluster was associated with higher average PPR output but lower consistency, suggesting a boom-or-bust profile. Another cluster showed more stable fantasy relevance with moderate production, while the third group captured players with generally lower output.

These findings highlight that rookie performance is not uniform across players. Instead, players tend to fall into different archetypes, which provides additional context beyond what is captured by supervised models. This type of analysis is useful for understanding risk profiles and variability in rookie outcomes, which can be valuable for fantasy decision-making.

F. Feature Importance

To better understand why certain models performed well, feature importance was extracted from the random forest models for both regression and classification tasks. Unlike linear models, which rely on coefficients that assume linear relationships, random forests provide a more flexible way to measure how much each feature contributes to prediction accuracy.

For the regression task, draft position emerged as the most influential variable by a significant margin. With an importance score of approximately 43%, it outweighed all other individual features. This result aligns with real-world expectations, as

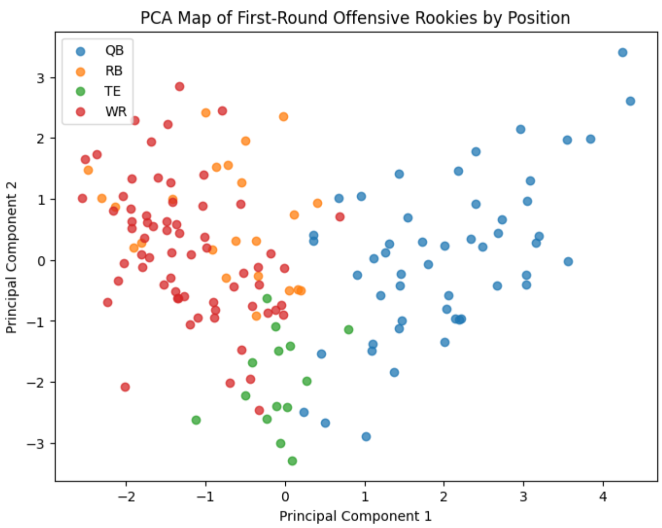

Fig. 2. PCA visualization of rookie players with KMeans clustering, showing natural groupings and positional separation

players selected earlier in the draft are more likely to receive immediate opportunities, leading to higher overall fantasy production.

Other features, including weight, college games played, and production metrics such as total yards and touchdowns, contributed moderately to the model. However, none approached the influence of draft capital. This suggests that opportunity, rather than just ability, plays a dominant role in determining total rookie output.

For the classification task, the importance distribution shifted. College production metrics, particularly total yards and touchdowns, became the most influential predictors of fantasy relevance. This indicates that while draft position influences opportunity, actual on-field production in college is more predictive of whether a player will deliver meaningful fantasy performance.

This difference between regression and classification tasks was one of the more insightful findings in the study. It highlights that different aspects of player evaluation matter depending on the prediction objective. Draft capital drives volume and opportunity, while production metrics better capture efficiency and readiness.

Feature importance analysis provides both validation and interpretability for the modeling results. It explains why tree-based models performed well and reinforces the idea that combining draft information with production data is critical when evaluating rookie players.

V. Results and Discussion

The following results focus on evaluating model performance across tasks rather than producing final predictive outputs.

A. Model Comparison

The differences between model performance became more meaningful once the results were interpreted in practical terms. For the classification task, Random Forest produced the strongest results with a ROC-AUC of approximately 66%. This means the model has about a 66% chance of correctly distinguishing between a fantasy-relevant player and a non-relevant player when comparing two random cases. While this is not extremely high, it does show that the model is learning useful patterns beyond random guessing.

Logistic Regression performed slightly lower at around 64%, while Decision Tree models were in a similar range. The key difference was consistency. Random Forest maintained a better balance between precision and recall, meaning it was more reliable at both identifying relevant players and avoiding false positives.

For the regression task, performance is best understood in terms of prediction error measured in fantasy points. Random Forest achieved an RMSE of approximately 94.3 points, compared to 97.2 for Linear Regression and 98.1 for Decision Tree Regression. In practical terms, this means the model's predictions are off by about 94 fantasy points on average when accounting for larger errors.

While a difference of 3 to 4 points in RMSE may seem small, it represents a noticeable improvement given the relatively small dataset and the wide range of rookie outcomes. In fantasy football terms, a difference of even 50 to 100 points over a season can separate starting-caliber players from bench-level contributors.

Tree-based models provided the most reliable and consistent results across both tasks. Random Forest, in particular, was able to capture nonlinear relationships and interactions between features that simpler models could not. This resulted in improved classification ability and reduced prediction error, making it the strongest overall model in this study.

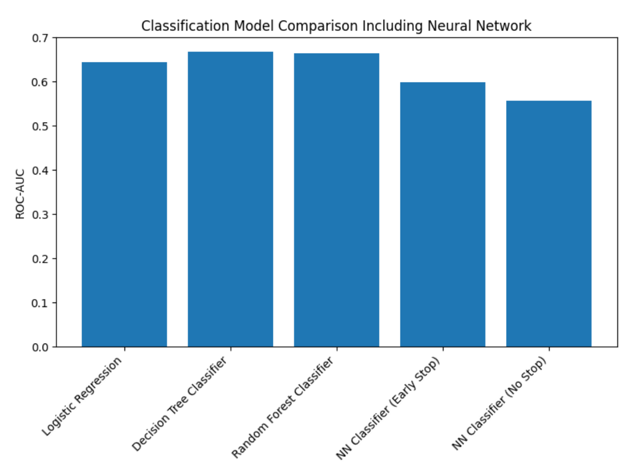

Fig. 3. Classification model comparison using ROC-AUC

B. Neural Network Behavior

Neural network performance ended up being one of the more revealing parts of this study, mainly because it highlighted how model complexity interacts with dataset size.

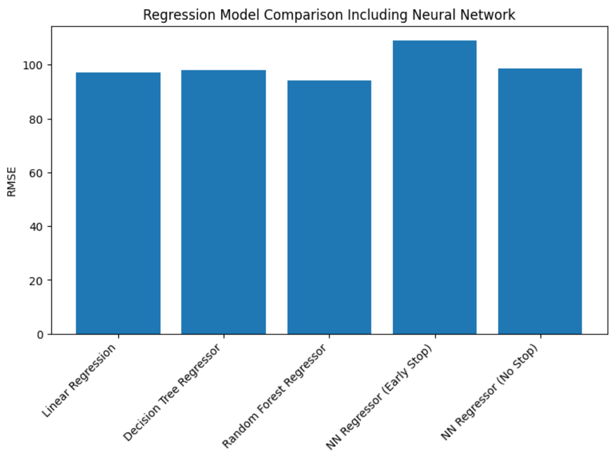

Fig. 4. Regression model comparison using RMSE

The model trained without early stopping showed clear signs of overfitting. Training loss continued to decrease steadily, while validation loss increased over time. In practical terms, this means the model was getting better at memorizing the training data but worse at making accurate predictions on new players.

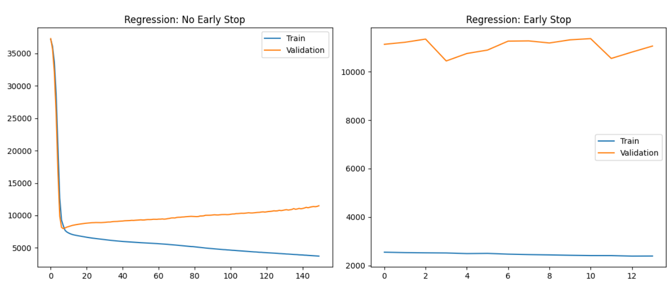

Fig. 5. Neural network training vs validation loss showing overfitting behavior without early stopping

When early stopping was applied, this behavior was reduced. Training stopped once validation performance stopped improving, which prevented the model from continuing to overfit. However, this came with a tradeoff. While the model became more stable during training, its overall predictive performance did not improve.

For regression, the neural network with early stopping produced an RMSE of approximately 109 points. This means predictions were off by about 109 fantasy points on average, which is significantly worse than Random Forest and even Linear Regression. The version without early stopping performed slightly better at around 98.7 points, but still did not outperform simpler models.

For classification, the neural network with early stopping achieved a ROC-AUC of approximately 60%, while the version without early stopping dropped to around 56%. This indicates that even in its more stable configuration, the model struggled to reliably separate fantasy-relevant players from non-relevant ones.

These results show that neural networks require more data to be effective. With only 164 players in the dataset, the model did not have enough information to learn generalizable patterns. Instead of improving performance, the added complexity introduced instability and overfitting, making simpler models the better choice for this problem.

C. Feature Importance

Feature importance analysis provided useful insight into why certain models performed better than others by showing which variables had the most influence on predictions.

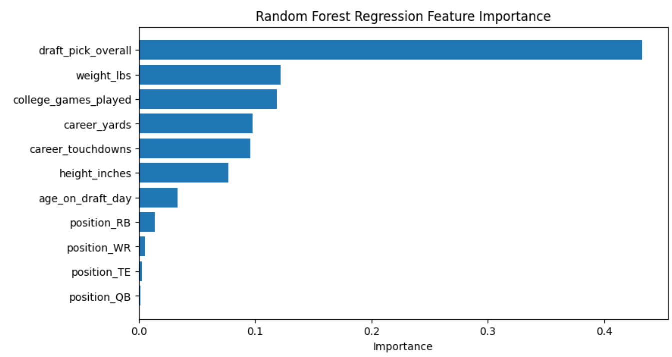

Fig. 6. Feature importance from the Random Forest regression model

For the regression task, draft position was by far the most important feature, contributing approximately 43% of the model's decision-making. This means nearly half of the model's ability to predict total fantasy points was driven by where a player was selected in the draft. In practical terms, earlier draft picks tend to receive more opportunities, which directly translates to higher fantasy production.

Other features such as weight, college games played, and college production metrics contributed at much lower levels, generally in the 7% to 12% range. While these variables still played a role, their impact was significantly smaller compared to draft capital.

For the classification task, the importance shifted away from draft position and toward college production. Total college yards contributed approximately 20% of the model's importance, while touchdowns contributed around 17%. This suggests that while draft position influences opportunity, actual production in college is a stronger indicator of whether a player will become fantasy relevant.

This difference between regression and classification is important. Draft capital is more useful for predicting total volume, while production metrics are better for identifying players who will deliver meaningful fantasy value. In other words, opportunity drives points, but performance drives relevance.

This analysis helps explain why Random Forest performed well. It was able to combine these different types of signals and capture how they interact, rather than relying on a single dominant factor.

D. Unsupervised Learning and Player Structure

Unsupervised learning provided additional context beyond prediction by showing how players naturally group based on their characteristics.

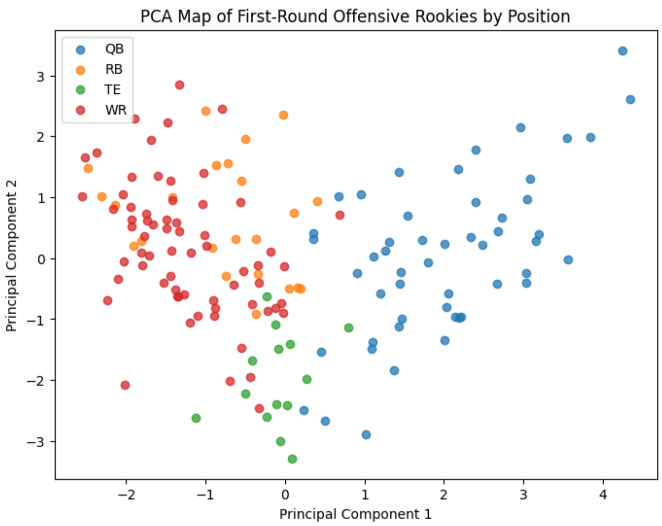

Fig. 7. PCA visualization of rookie players with KMeans clustering, showing natural groupings and positional separation

The PCA visualization revealed a clear separation between positions, with quarterbacks forming a distinct cluster compared to other skill positions. This indicates that the features used in the model, such as physical attributes and production metrics, capture meaningful differences between positions.

When looking at the plot, quarterbacks tend to occupy a different region of the feature space, while running backs, wide receivers, and tight ends show more overlap. This makes sense from a football perspective, as quarterbacks have fundamentally different roles and statistical profiles compared to other positions.

KMeans clustering further revealed three distinct groups of players. One cluster had higher average PPR production at around 191 points but a low fantasy relevance rate of approximately 14%. This suggests a boom-or-bust group, where a few players produce at a high level while most fail to become consistent contributors.

Another cluster showed more moderate production but a higher consistency in fantasy relevance, indicating a more stable group of players who are more likely to contribute at a usable level. The third cluster captured players with generally lower production across the board.

In practical terms, this means rookie outcomes are not evenly distributed. Instead, players tend to fall into different archetypes, such as high-risk, high-reward players or more consistent but lower-ceiling options. This type of insight is useful for fantasy decision-making because it adds context beyond just point predictions.

E. 2026 Rookie Class Predictions

To extend the analysis beyond model evaluation, the best-performing model, Random Forest, was used to generate forward-looking predictions for the 2026 rookie class.

TABLE I
TOP PREDICTED 2026 ROOKIE PERFORMERS

Player	Pos	Pred PPR	Relevant	Prob (%)
Jeremiyah Love	RB	279.8	Yes	71.8
Fernando Mendoza	QB	223.4	No	6.2
Carnell Tate	WR	197.5	Yes	52.6
KC Concepcion	WR	190.3	No	46.4
Ty Simpson	QB	170.7	No	39.8

The model produced two outputs for each player: predicted total rookie season PPR fantasy points and the probability of being classified as fantasy relevant. These predictions provide a practical application of the modeling pipeline and illustrate how historical patterns translate to future projections.

Among the 2026 class, the most interesting result comes from the contrast between projected production and fantasy relevance. Jeremiyah Love (RB) stands out as the top overall projection at approximately 280 PPR points with a 71.8% probability of being fantasy relevant, which aligns with expectations for a high draft pick with strong production.

However, the more surprising case is Fernando Mendoza (QB), who was selected first overall and projected for over 220 PPR points, yet only received a 6.2% probability of being fantasy relevant. This highlights a key limitation in evaluating quarterbacks for fantasy purposes. While total point production may appear strong, the positional threshold for relevance is much stricter, making it significantly harder to classify as a top performer.

The remaining players fall into a more uncertain range. Carnell Tate (WR) sits near the threshold with a 52.6% probability, suggesting a true borderline outcome that likely depends on team context and opportunity. Other players in the class cluster below the 50% mark, reinforcing how difficult it is for rookies to immediately produce consistent fantasy value.

Rather than producing confident predictions across the board, the model outputs a range of probabilities that reflect the inherent uncertainty in rookie performance. This is actually a desirable outcome, as it indicates the model is capturing variability instead of forcing overly confident conclusions.

F. Personal Interpretation

When stepping back across all phases of this study, a few consistent patterns stand out that help explain both model performance and rookie fantasy outcomes.

The most important takeaway is that model simplicity outperformed model complexity in this setting. While neural networks provided a flexible framework, they were unable to generalize effectively due to the limited dataset size. In contrast, Random Forest models consistently produced the strongest results across both regression and classification tasks. This reinforces an important principle in statistical learning, that model selection should be driven by the structure and size of the data rather than the perceived power of the model.

Another key finding is the distinct role that different features play depending on the prediction objective. Draft capital emerged as the dominant factor in predicting total fantasy production, accounting for a large portion of the model's decision-making. This reflects the real-world importance of opportunity, as players selected earlier in the draft are more likely to receive immediate playing time and usage. However, when shifting to classification, college production metrics such as yards and touchdowns became more influential. This suggests that while opportunity drives volume, performance efficiency is what ultimately determines whether a player becomes fantasy relevant.

The prediction phase further reinforced these findings by applying the trained model to the 2026 rookie class. The results highlighted meaningful distinctions between total projected production and fantasy relevance classification. For example, certain players were projected to produce relatively high PPR point totals but were still classified as not fantasy relevant due to positional thresholds. This demonstrates that high production alone does not guarantee meaningful fantasy value, particularly at positions with stricter ranking requirements such as quarterback.

Additionally, the model's probability outputs provided insight into confidence levels, rather than forcing binary decisions. Many players fell into intermediate probability ranges, reflecting the inherent uncertainty of rookie performance. This behavior suggests that the model is capturing realistic variability rather than overfitting to deterministic patterns.

Unsupervised learning results added another layer of understanding by showing that players naturally group into different archetypes. The PCA and clustering analysis revealed that rookies do not follow a single trajectory, but instead fall into categories such as high-variance, high-upside players and more consistent but lower-ceiling contributors. This helps explain why prediction error remains relatively high, as the outcome distribution itself is highly variable.

This study demonstrates that rookie performance is driven by a combination of opportunity and prior production, with each factor contributing differently depending on the modeling task. More importantly, it shows that even with relatively simple features, machine learning models can extract meaningful patterns that provide value for fantasy decision-making.

Rather than producing a final deterministic prediction system, the primary contribution of this work is in evaluating how different models behave, identifying which approaches generalize best, and demonstrating how these models can be applied in a practical context. This foundation provides a clear path for future improvements while already offering actionable insights for evaluating incoming rookie classes.

VI. Conclusion

Going into this project, I expected more complex models like neural networks to perform better given their flexibility. What stood out most from the results was that this was not the case. Random Forest consistently produced better and more stable performance across both regression and classification tasks, even though it is a simpler model compared to neural networks.

This ended up being less about model capability and more about the structure of the data. With a relatively small dataset and highly structured features, the added complexity of neural networks led to overfitting rather than better generalization. In contrast, tree-based models were better suited to capture nonlinear relationships without requiring large amounts of data.

Another takeaway was how differently features behave depending on the prediction goal. Draft capital clearly drives total production, but it does not guarantee fantasy relevance. College production metrics, on the other hand, played a bigger role in identifying players who actually deliver usable fantasy value. That distinction became much clearer after seeing the classification results alongside the regression outputs.

The prediction results for the 2026 class reinforced this idea even further. Some players projected for relatively high point totals were still unlikely to be fantasy relevant based on positional thresholds. That disconnect highlights how raw production alone does not fully translate to fantasy value.

If I were to approach this again, I would focus earlier on expanding the dataset and incorporating team-level context, since opportunity appears to be one of the most important drivers of rookie outcomes. I would also be more cautious about introducing complex models without first considering whether the data can actually support them.

This project showed that even with a relatively simple feature set, machine learning models can capture meaningful patterns in rookie performance. More importantly, it showed that understanding model behavior is just as important as the predictions themselves when applying these methods in practice.

VII. Future Work

While this study successfully evaluates multiple statistical learning approaches and produces actionable rookie predictions, several opportunities exist to extend and improve the modeling framework.

One of the most important areas for future work is expanding the dataset to better align with the full scope originally outlined for this project. The current dataset primarily includes player attributes, draft capital, and college production metrics. However, a more complete modeling approach would also incorporate team context, coaching tendencies, offensive environment, depth chart competition, and engineered opportunity metrics. Incorporating these additional variables would allow the model to better capture real-world factors that significantly influence rookie performance.

In particular, features such as vacated targets, team scoring environment, quarterback quality, and competition for touches could provide meaningful improvements in predictive accuracy. These additions would help reduce the gap between statistical projections and actual fantasy outcomes by modeling opportunity more explicitly. With a richer feature space and an expanded dataset size, it is expected that both RMSE and MAE would decrease, resulting in more precise rookie point predictions.

Another key area for improvement is the application of reinforcement learning techniques. While the current models operate in a static supervised learning framework, reinforcement learning offers the potential to continuously update and refine predictions as new information becomes available. For example, a reinforcement learning agent could adjust player valuations based on simulated draft strategies, roster construction, or evolving player roles over time.

This type of adaptive learning framework would be particularly valuable for dynasty fantasy football, where long-term player value and uncertainty play a major role in decision-making. By incorporating feedback loops and sequential decision-making, reinforcement learning could help optimize draft strategies and improve prediction accuracy beyond what is possible with static models.

Additionally, expanding the dataset to include a larger number of rookie classes would improve the performance of more complex models such as neural networks. As observed in this study, neural networks struggled due to the relatively small sample size of approximately 164 players. Increasing the dataset toward the projected 200–400 player range would provide the necessary scale for these models to learn more generalizable patterns.

Looking ahead, the goal is to refine this modeling framework into a practical tool for fantasy football decision-making. With continued dataset expansion, improved feature engineering, and the integration of adaptive learning techniques, the model can be used to support dynasty rookie drafts, including the upcoming 2026 fantasy draft cycle. Ultimately, this would allow for more informed, data-driven decisions that complement traditional scouting and analysis.

REFERENCES

[1] Pro Football Reference, "NFL Player Statistics and Draft Data." [Online]. Available: https://www.pro-football-reference.com/

[2] G. James, D. Witten, T. Hastie, and R. Tibshirani, *An Introduction to Statistical Learning with Applications in Python*. Springer, 2023.

[3] Scikit-learn Developers, "Scikit-learn: Machine Learning in Python." [Online]. Available: https://scikit-learn.org/

[4] TensorFlow Developers, "TensorFlow: Large-Scale Machine Learning Platform." [Online]. Available: https://www.tensorflow.org/

[5] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.